

- 5 A manager claims that the lengths of the rubber tubes that his company produces have a median of 5.50 cm. The lengths, in cm, of a random sample of 11 tubes produced by this company are as follows.

5.56    5.45    5.47    5.58    5.54    5.52    5.60    5.35    5.59    5.51    5.62

It is required to test at the 10% significance level the null hypothesis that the population median length is 5.50 cm against the alternative hypothesis that the population median length is not equal to 5.50 cm.

Show that both a sign test and a Wilcoxon signed-rank test give the same conclusion and state this conclusion. [9]

[illegible]